SPECIFICATION

1.77” TFT display
No Touch Panel

- Active Matrix TFT LCD Module
- 128 x RGB x 160 Dots
- 65K Color
- Interfaces: 8/9/16/18 bit MCU
  16/18 bit RGB
  3/4 wire SPI
- Connection Type: FPC Socket

Part Number: FDS128x160(28.02x35.04)RGE-17N

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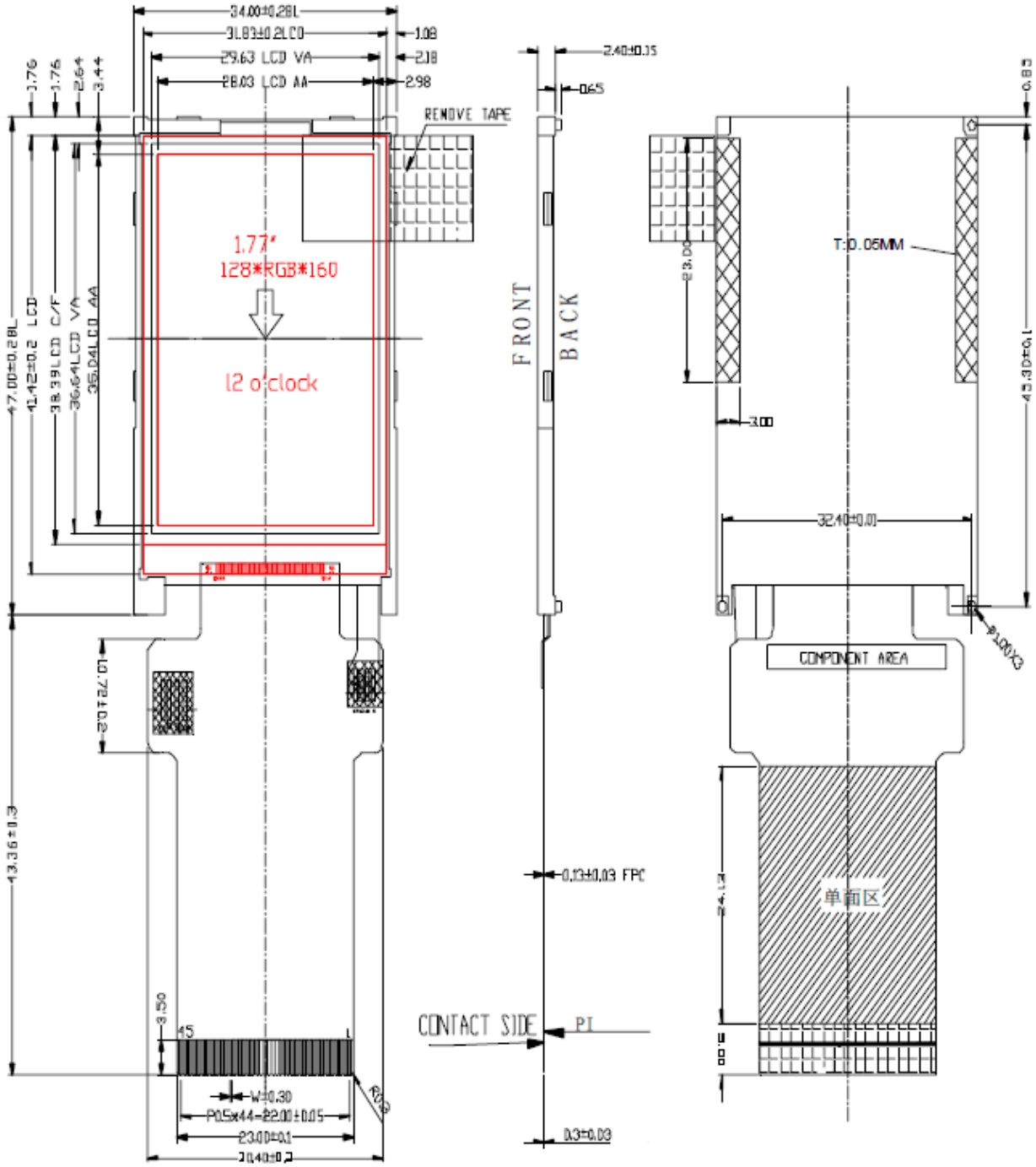

Features:

1. **1.77" TFT**-LCD, 65K COLOR
2. DRIVER IC: ILI9163V
3. OP: -20°C~70°C / ST: -30°C~80°C
4. BACKLIGHT: LED
5. ROHS Compliant

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GENERAL DESCRIPTION

- 1.77” (diagonal), 128xRGBx160 QQVGA,
- Transmissive/Normally White TFT module
- Viewing Direction: 6 o’clock
- Driving IC: ILI9163V or equivalent
- 300 NITS
- 8/9/16/18 bit MCU, 3/4 wire SPI, 16/18 bit RGB interface options

General Specifications

PARAMETER	SPECIFICATIONS	UNIT
Outline dimensions	35.6(W) x 47(H) x 3.5(D) (Exclude FPC, cables of backlight)	mm
View area	31.88(W) x 38.39(H)	mm
LCD active area	28.03(W) x 35.04(H)	mm
Driver element	TFT Active Matrix	-
Pixel arrangement	RGB Vertical Stripe	-
Pixel pitch	0.219(W) x 0.219(H)	mm

ELECTRICAL Maximum Rating

ITEM	SYMBOL	MIN	MAX	UNIT	NOTE
Digital Supply voltage	V_{DD}	-0.3	4.8	V	
Digital Interface Supply voltage	V_{DDIO}	-0.3	4.6	V	

ELECTRICAL Specifications

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT	REMARK
Digital supply voltage	V_{DD}	2.5	3.3	4.2	V	
Digital interface supply voltage	V_{DDIO}	1.65	3.3	4.2	V	
Normal mode current consumption	I_{DD}	--	1.5	--	mA	
Level input voltage	V_{IH}	$0.7V_{DDIO}$		V_{DDIO}	V	
	V_{IL}	GND		$0.3V_{DDIO}$	V	
Level output voltage	V_{OH}	$0.8V_{DDIO}$		V_{DDIO}	V	
	V_{OL}	GND		$0.2V_{DDIO}$	V	

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Interface Pin Connections

<table>
<tr><th>PIN NO.</th><th>SYMBOL</th><th>DESCRIPTION</th><th>I/O</th></tr>
<tr><td>1</td><td>GND</td><td>Ground</td><td>P</td></tr>
<tr><td>2</td><td>LEDK</td><td>Cathode pin of backlight</td><td>P</td></tr>
<tr><td>3</td><td>LEDA1</td><td>Anode pin of backlight</td><td>P</td></tr>
<tr><td>4</td><td>LEDA2</td><td>Anode pin of backlight</td><td>P</td></tr>
<tr><td>5</td><td>LEDA3</td><td>Anode pin of backlight</td><td>P</td></tr>
<tr><td>6</td><td>IM0</td><td rowspan="2">MCU parallel interface type selection
<table>
<tr><th>IM1</th><th>IM0</th><th>Parallel interface</th></tr>
<tr><td>0</td><td>0</td><td>MCU 8-bit Parallel</td></tr>
<tr><td>0</td><td>1</td><td>MCU 16-bit Parallel</td></tr>
<tr><td>1</td><td>0</td><td>MCU 9-bit Parallel</td></tr>
<tr><td>1</td><td>1</td><td>MCU 18-bit Parallel</td></tr>
</table></td><td rowspan="2">I</td></tr>
<tr><td>7</td><td>IM1</td></tr>
<tr><td>8</td><td>IM2</td><td>MCU Parallel interface bus and Serial interface select
—IM2=’1’; Parallel interface
—IM2=’0’; Serial interface</td><td>I</td></tr>
<tr><td>9</td><td>RCM1</td><td>RGB and MCU interface mode selection pin.
—RCM1=1, RGB interface
—RCM1=0, MCU interface mode</td><td>I</td></tr>
<tr><td>10</td><td>SPI4W</td><td>SPI interface selection pin
SPI4W=’0’: 3-wire SPI
SPI4W=’1’: 4-wire SPI
This pin is internal pull low.</td><td>I</td></tr>
<tr><td>11</td><td>VCC</td><td>Supply voltage (3.3V)</td><td>P</td></tr>
<tr><td>12</td><td>VCC</td><td>Supply voltage (3.3V)</td><td>P</td></tr>
<tr><td>13</td><td>SDI</td><td>When RCM1, RCM0=’1X’ (RGB I/F), serial input/output signal in serial I/F mode. The data is in put on the rising edge of the SCL signal. The data is output on the falling edge of the SCL signal.
When RCM1, RCM0=’0X’ (MCU I/F), this pin is not used, and fix at VDDI or GND level.
If not used, please fix this pin at VDDI or GND level.</td><td>I</td></tr>
<tr><td>14</td><td>RD</td><td>Read enable
If not used, please connect this pin to VDD.</td><td>I</td></tr>
<tr><td>15</td><td>DC/SCL</td><td>Display data / Command selection pin in parallel and SCL in 3-pin SPI interface.
D/CX=’1’: Display data
D/CX=’0’: Command data
If not used, please connect this pin to GND.</td><td>I</td></tr>
</table>

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Interface Pin Connections *(continued)*

PIN NO.	SYMBOL	DESCRIPTION	I/O
16	WR/SPI_RS	Write enable in parallel interface. WRX: for 8080 MCU D/CX: for 4-wire SPI If not used, please connect this pin to VDDI or GND.	I
17	CS	Chip select input pin (“Low” enable). This pin can be permanently fixed “Low” in MCU interface mode only.	I
18	RESET	This signal will reset the device and must be applied to properly initialize the chip.	I
19-36	DB17-DB0(SDA)	When RCM1=’0’ (MCU I/F), D[17:0] are used to MCU parallel interface data bus, and D0 is also the serial input / output signal in SPI interface mode. In serial interface, D[17:0] are used to RGB interface data bus.	I/O
37	PCLK	Pixel clock signal in RGB I/F mode. If not used, please fix this pin at GND level.	I
38	DE	Data enable signal in RGB I/F mode. If not used, please fix this pin at GND level.	I
39	HSYNC	Horizontal sync. Signal in RGB I/F mode. If not used, please fix this pin at GND level.	I
40	VSYNC	Vertical sync. Signal in RGB I/F mode. If not used, please fix this pin at GND level.	I
41	XR(NC)		
42	YD(NC)		
43	XL(NC)		
44	YU(NC)		
45	GND	Ground	P

Environmental Conditions

ITEM	Operating Temperature (Topr)		Storage Temperature (Tstg)		REMARK
	MIN	MAX	MIN	MAX	
Operating Temperature	-20°C	+70°C	-30°C	+80°C	Dry

Note 1: Product cannot sustain at extreme storage conditions for long time.

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